ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Acids, Bases and Salts** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 10	Bases turn red litmus blue and have pH above 7.
2	**(b)** 1.4	Stomach has hydrochloric acid.
3	**(b)** magnesium hydroxide	$Mg(OH)_2$ neutralises acid.
4	**(c)** HCl	Egg-shell is $CaCO_3$; acids release CO_2.
5	**(c)** 5.5	Enamel dissolves below pH 5.5.
6	**(d)** basic solution	Pink in bases, colourless in acids.
7	**(c)** 16 mL	Proportion: 8 x 20/10 = 16 mL.
8	**(b)** tartaric acid	Tamarind contains tartaric acid.
9	**(d)** red	Red in acid, yellow in base.
10	**(c)** antacid	Antacids neutralise excess stomach acid.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Acids give H^+ ions only in the presence of water (1); Dry HCl has no H^+ ions free (1).	1+1
12	Acid forms H^+ (hydronium) ions in water (1); Moving ions carry current (1).	1+1
13	Calcium oxychloride (1); $CaOCl_2$ (1).	1+1
14	Dilution is highly exothermic (1); Adding water to acid may splash hot acid and cause burns (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Acid + metal -> salt + hydrogen (1); $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (1); Hydrogen burns with a pop sound (1).	1+1+1
16	Electrolysis of brine (aqueous NaCl) (1); $2\,NaCl + 2\,H_2O \longrightarrow 2\,NaOH + Cl_2 + H_2$ (1); Products: NaOH, chlorine, hydrogen (1).	1+1+1
17	$HCl + H_2O \longrightarrow H_3O^+ + Cl^-$ (1); Free H^+ cannot exist alone; it combines with water (1); H_3O^+ is the hydronium ion (1).	1+1+1
	OR: Salt, water and carbon dioxide are formed (1); $Na_2CO_3 + 2\,HCl \longrightarrow 2\,NaCl + H_2O + CO_2$ (1); CO_2 turns lime water milky (1).	1+1+1
18	$Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O$ (1); Used for bleaching cotton/linen (1); Used to disinfect drinking water (1).	1+1+1
	OR: A is more acidic (1); pH difference is 3 (1); Concentration differs by 1000 times (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Bubbles on the surface of zinc (1); $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (1); Gas burns with a pop sound near a flame (1); Test tube becomes warm (exothermic) (1); Acid + metal -> salt + hydrogen (1).	1+1+1+1+1
	OR: Electrolysis of brine (1); $2\,NaCl + 2\,H_2O \longrightarrow 2\,NaOH + Cl_2 + H_2$ (1); Cl2 at anode, H2 at cathode (1); Uses: NaOH in soap; Cl_2 in water treatment (1); H_2 as fuel/ margarine (1).	1+1+1+1+1
20	Set up circuit with bulb and two electrodes (1); HCl solution: bulb glows (1); Glucose solution: bulb does not glow (1); Acids give H^+ ions in water (1); Glucose does not give ions (1).	1+1+1+1+1
	OR: $Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O$ (1); Bleaching powder: bleaching, disinfecting water (1); $NaCl + H_2O + CO_2 + NH_3 \longrightarrow NH_4Cl + NaHCO_3$ (1); Baking soda: baking powder, antacid (1); Soda-acid extinguisher also uses it (1).	1+1+1+1+1

Mistakes to avoid:

- Pouring water into concentrated acid; always add the acid slowly to water with stirring because the process is highly exothermic.
- Writing that a solution with pH 3 is 'less acidic' than one with pH 5; a lower pH means a **stronger** acid.
- Mixing up the formulae: baking soda is $NaHCO_3$, washing soda is $Na_2CO_3 \cdot 10\,H_2O$, plaster of Paris is $CaSO_4 \cdot \frac{1}{2}\,H_2O$ and bleaching powder is $CaOCl_2$.

- Thinking that all compounds containing hydrogen are acids; glucose and alcohol do not release $\mathrm{H^+}$ ions.
- Writing $\mathrm{H_2}$ as the gas formed when an acid reacts with a carbonate; the gas is $\mathrm{CO_2}$.
- Assuming neutralisation always gives a neutral solution; a strong acid with a weak base gives an acidic salt solution (and vice-versa).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Acids, Bases and Salts** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** methanoic acid	Methanoic acid causes the burning pain.
2	**(b)** 7	Water is neutral.
3	**(d)** it absorbs water and sets to gypsum	$CaSO_4 \cdot \frac{1}{2} H_2O$ reacts with water to form gypsum.
4	**(c)** chlorine	$2\,Cl^- \longrightarrow Cl_2$ at anode.
5	**(a)** 5.6	Rain is normally slightly acidic; below 5.6 is acid rain.
6	**(c)** baking soda	$NaHCO_3$ is anhydrous.
7	**(b)** dry slaked lime	$Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O$.
8	**(b)** it has no H^+ ions without water	Acids show acidic behaviour only in water.
9	**(c)** turns lime water milky	The gas is CO_2.
10	**(c)** tartaric acid	Acid releases CO_2 with baking soda.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	HCl and HNO_3 ionise to give H^+ ions in water (1); Alcohol and glucose contain hydrogen but do not give H^+ (1).	1+1
12	It absorbs moisture to form gypsum (1); $CaSO_4 \cdot \frac{1}{2} H_2O + 1.5\,H_2O \longrightarrow CaSO_4 \cdot 2\,H_2O$; it hardens and is no longer useful (1).	1+1
13	$2\,NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$ (2).	2
14	Distilled water has no ions (1); Rain water has dissolved acids/salts giving ions (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Acid attacks the calcium phosphate enamel (1); Bacteria produce acid from sugar (1); Toothpaste is basic and neutralises the acid (1).	1+1+1
16	Heat baking soda to get sodium carbonate; recrystallise (1); $Na_2CO_3 + 10\,H_2O \longrightarrow Na_2CO_3 \cdot 10\,H_2O$ (1); Uses: glass, soap, paper, softening hard water (1).	1+1+1
17	It neutralises excess stomach acid (1); $Mg(OH)_2 + 2\,HCl \longrightarrow MgCl_2 + 2\,H_2O$ (1); It is a mild base, so safe (1).	1+1+1
	OR: Acidic behaviour is due to H^+/H_3O^+ (1); H^+ forms only when acid dissolves in water (1); Dry HCl gas has no free ions (1).	1+1+1
18	Calcium sulphate hemihydrate $CaSO_4 \cdot \frac{1}{2} H_2O$ (1); Heating gypsum at 373 K (1); Used for plaster of fractured bones and toys (1).	1+1+1
	OR: Baking soda reacts with an acid (1); $NaHCO_3 + H^+ \longrightarrow Na^+ + H_2O + CO_2$ (1); CO_2 smothers the fire (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$2\,NaOH + Zn \longrightarrow Na_2ZnO_2 + H_2$ (1); $Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$ (1); Non-metal oxides are acidic (1); Metal oxides are basic (1); Base + acid oxide gives salt and water (1).	1+1+1+1+1
	OR: $2\,NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$, then recrystallisation (1); Washing soda: glass, soap, softening hard water (1); $CaSO_4 \cdot 2\,H_2O \longrightarrow CaSO_4 \cdot \frac{1}{2} H_2O + 1.5\,H_2O$ at 373 K (1); POP: plaster for fractured bones (1); POP: toys, decoration, smoothening surfaces (1).	1+1+1+1+1
20	$Na_2CO_3 + 2\,HCl \longrightarrow 2\,NaCl + H_2O + CO_2$ (1); $NaHCO_3 + HCl \longrightarrow NaCl + H_2O + CO_2$ (1); $Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$: milky (1); $CaCO_3 + H_2O + CO_2 \longrightarrow Ca(HCO_3)_2$ with excess (1); Milkiness disappears on excess CO_2 (1).	1+1+1+1+1
	OR: Fixed water molecules in the crystal of salt (1); Blue crystals turn white on heating (1); Water is lost (1); Add water: blue colour returns (1); Examples: $Na_2CO_3 \cdot 10\,H_2O$, $CaSO_4 \cdot 2\,H_2O$ (1).	1+1+1+1+1

Mistakes to avoid:

- Pouring water into concentrated acid; always add the acid slowly to water with stirring because the process is highly exothermic.
- Writing that a solution with pH 3 is 'less acidic' than one with pH 5; a lower pH means a **stronger** acid.
- Mixing up the formulae: baking soda is $\mathrm{NaHCO_3}$, washing soda is $\mathrm{Na_2CO_3 \cdot 10\,H_2O}$, plaster of Paris is $\mathrm{CaSO_4 \cdot \tfrac{1}{2}\,H_2O}$ and bleaching powder is $\mathrm{CaOCl_2}$.
- Thinking that all compounds containing hydrogen are acids; glucose and alcohol do not release $\mathrm{H^+}$ ions.
- Writing $\mathrm{H_2}$ as the gas formed when an acid reacts with a carbonate; the gas is $\mathrm{CO_2}$.
- Assuming neutralisation always gives a neutral solution; a strong acid with a weak base gives an acidic salt solution (and vice-versa).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Acids, Bases and Salts** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** add acid slowly to water	Dilution is highly exothermic.
2	**(a)** HCl	HCl is a strong acid, giving more H^+.
3	**(c)** 5	$CuSO_4 \cdot 5\,H_2O$.
4	**(a)** CO_2	$CaCO_3$ is formed.
5	**(c)** decrease	Lactic acid forms.
6	**(a)** white	Anhydrous $CuSO_4$ is white.
7	**(a)** neutralisation	Standard definition.
8	**(b)** $CaOCl_2$	Calcium oxychloride.
9	**(b)** $Na_2CO_3 \cdot 10\,H_2O$	Sodium carbonate decahydrate.
10	**(b)** plaster of Paris	$CaSO_4 \cdot 2\,H_2O \longrightarrow CaSO_4 \cdot \frac{1}{2}\,H_2O$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Fixed number of water molecules in one formula unit of a salt crystal (1); $CuSO_4 \cdot 5\,H_2O$ (1).	1+1
12	Reaction of an acid with a base to give salt and water (1); $NaOH + HCl \longrightarrow NaCl + H_2O$ (1).	1+1
13	Methanoic (formic) acid (1); Rub baking soda or a mild base (1).	1+1
14	They contain acids that react with the metals (1); Harmful poisonous compounds may form and spoil the food (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Alkaline milk does not turn sour easily (1); Lactic acid formed is neutralised first (1); So it takes a longer time to set as curd (1).	1+1+1
16	Na_2CO_3, water and CO_2 (1); Uses: baking powder, antacid (1); Soda-acid fire extinguisher (1).	1+1+1
17	pH measures H^+ concentration on a scale of 0 to 14 (1); Lemon juice about 2.2; blood about 7.4 (1); Pure water 7 (1).	1+1+1
	OR: $2\,NaOH + Zn \longrightarrow Na_2ZnO_2 + H_2$ (1); $CaCO_3 + 2\,HCl \longrightarrow CaCl_2 + H_2O + CO_2$ (1); $2\,NaOH + CO_2 \longrightarrow Na_2CO_3 + H_2O$ (1).	1+1+1
18	NaCl is neutral (1); NH_4Cl is acidic (1); CH_3COONa and Na_2CO_3 are basic (1).	1+1+1
	OR: A neutral, B and D acidic, C basic (1); Increasing H^+: C < A < D < B (1); Lower pH means more H^+ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$NaOH + HCl \longrightarrow NaCl + H_2O$ (1); $Mg(OH)_2 + 2\,HCl \longrightarrow MgCl_2 + 2\,H_2O$ (1); $CuO + 2\,HCl \longrightarrow CuCl_2 + H_2O$ (metal oxide) (1); $CO_2 + 2\,NaOH \longrightarrow Na_2CO_3 + H_2O$ (1); Salt and water formed in all cases (1).	1+1+1+1+1
	OR: Acid rain (pH below 5.6) lowers river pH (1); Aquatic life is endangered (1); Bee/ant sting has methanoic acid (1); Nettle leaves inject methanoic acid (1); Treated by a mild base like baking soda (1).	1+1+1+1+1
20	Scale from 0 to 14 measuring H^+ concentration (1); pH 7 neutral, below 7 acidic, above 7 basic (1); Stomach acid (pH about 1.4) helps digestion (1); Tooth enamel is attacked below pH 5.5 (1); Plants grow best in soil of a particular pH (1).	1+1+1+1+1
	OR: Litmus: blue to red in acid; red to blue in base (1); Phenolphthalein: colourless in acid, pink in base (1); Methyl orange: red in acid, yellow in base (1); Olfactory indicators: onion, vanilla, clove (1); Smell lost in base for onion and vanilla; clove retains it in acid (1).	1+1+1+1+1

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